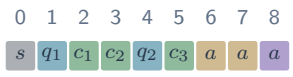
$$M = \mathrm{LN}(H)W_{mem}^{\mathsf{T}}, \quad O = \mathrm{Route}^2(O^0; M), \quad F = \mathrm{Decoder}^4(F^0; M)$$
$$O^0 = CW_c^{\mathsf{T}} + LW_l^{\mathsf{T}} + (UW_{oq}^{\mathsf{T}})_{owner}$$

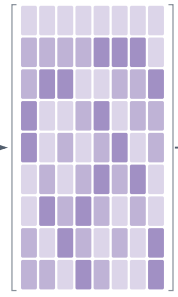
1. Token layout



x
T integer token IDs

2. Clef

64 causal hybrid blocks
d=5120; joint width w=1024
all fields in one sequence



H
$T \times d$

LN_d
Linear
$d \to w$

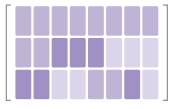


M
$T \times w$

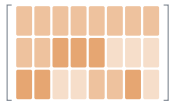
3. Contextual states and indices

$H_c = LN_d(H)$; $g = (H_c)_{-1}$
C,U: span means of H_c
M: all valid token memory

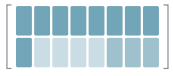
4. Context, lexical and question features



C
$N \times d$
mean option spans



L
$N \times d$
mean LM output rows



U
$Q \times d$
mean instruction spans

3 Linear $d \to w$
owner
broadcast; sum



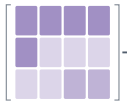
O^0
$N \times w$

2× evidence layer
$\mathrm{CA}(LN(O), LN(M))$
residual; pre-LN
FFN + residual



O
$N \times w$

5. Candidate-weighted fields, then global field interaction



O
$N \times w$

$$F_{base} = UW_{field}^{\mathsf{T}}$$
$$\alpha = \mathrm{softmax}_K(O \cdot F_{base}/\sqrt{w})$$
$$F^0 = F_{base} + LN(\textstyle\sum \alpha O)$$
$$+W_g g + e_{type}$$



F^0
$Q \times w$

4× TransformerDecoder
field self-attn;
cross-attn to M
FFN; residuals; final LN



F
$Q \times w$

Global field visibility
full token-memory access
companion scorer
plate follows

Axes $d = 5120$, $w = 1024$; One record, Q questions, N total options (N is not a fixed K per field). The illustration uses three option rows and two fields; w=1024.

Objects Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.

Mechanism Two option-to-memory cross-attention layers, then four field decoder layers. Fields communicate globally; lexical prior plus a gated cosine/MLP term gives scores.

Released source exposes inference and head configuration, not the training loss or historical optimization. Model-card base name and config class are recorded separately.